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Epistemic Limits and Formal Logic:

Learning About the Richness of Traditional Ecological Knowledge Through Artificial Intelligence

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Abstract

This thesis investigates the epistemological and technical limits of Artificial Intelligence (AI) when applied to the rich, context-dependent domain of Sámi Traditional Ecological Knowledge (TEK). Motivated by the escalating threats to socio-ecological endurance in Sápmi and the risks of "hermeneutical injustice" posed by statistical AI models, this research employs a constructive design approach. By attempting to encode the narrative practice of the "Sunrise Pasture Assessment" into the formal logic of Social Practice Theory (SPT), the study performs a boundary analysis, using the rigidity of code to highlight precisely what resists digitization.

The formalization demonstrates that while SPT can successfully coordinate the high-level syntax of herding, such as roles, resources, and conditional heuristics, it encounters hard structural boundaries when attempting to encapsulate the embodied, relational semantics of the practice. The analysis identifies key epistemic gaps, including ontological blindness to unencoded sensory variables, the erasure of granular herd dynamics, and the inability to process the historical continuity of the Sámi landscape.

This thesis reframes digitization bias as possibilities for strategic abstraction. It argues for a variable-resolution approach to AI design, where systems explicitly manage structural landmarks while deferring execution to human epistemic authority. Ultimately, delineating these computational boundaries serves as a mechanism of protection against technological solutionism. It resolves the paradox of participation by proving that while AI may coordinate logistics, the core wisdom of Sápmi resides strictly within the Sámi community and must remain sovereign.

Keywords: *Traditional Ecological Knowledge, Social Practice Theory, Knowledge Representation, Responsible AI, Sámi Reindeer Herding, Epistemic Justice.*

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Note on Terminology

Throughout this thesis, specific Northern Sámi concepts such as *guohtun* (conditions for grazing) and *siida* (the herding community unit) are retained in their original language rather than translated into English approximations like "pasture" or "village." This decision is intentional. As this research demonstrates, these terms encapsulate complex ecological and social relations that Western definitions often obscure. Preserving the original terminology serves as a reminder of the epistemic gaps that this thesis seeks to highlight: the nuances that resist simple translation into both English and code.

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1 Introduction

The Arctic acts as a major temperature regulator for the Earth [1]. As the Earth's climate has changed due to anthropogenic effects, Arctic sea ice extent has decreased drastically. This loss constitutes one of the most severe positive feedback loops in the climate system; the open ocean absorbs sunlight that sea ice would otherwise reflect, thereby accelerating global temperature increases. Indigenous populations in the Arctic face disproportionate impacts from these shifts, encountering fundamental ecological, cultural, and livelihood disruptions [2].

The Sámi people of Sápmi, spanning northern Norway, Sweden, Finland, and parts of Russia, maintain a way of life intrinsically linked to the Arctic environment through practices such as reindeer husbandry and fishing. Central to this relationship is Traditional Ecological Knowledge (TEK), a cumulative body of knowledge, practices, and beliefs developed by Indigenous and local cultures over centuries of environmental interaction [3]. TEK encompasses deep understandings of ecological relationships, seasonal cycles, and sustainable resource use, passed through generations via cultural transmission. Inherently place-based and dynamic, TEK offers a distinct component for reimagining scientific approaches to climate sustainability.

The Arctic is experiencing climate change at nearly four times the global average rate, leading to significant ecological and cultural consequences. Besides losing the ability to regulate the temperature of the earth, this accelerated warming also has heavy and unequal consequences for everyone who call the Arctic home, such as the Sámi, which are among those most directly affected [4]. Concurrently, Western science provides predictive capabilities through climate modelling and environmental monitoring that can adapt rapidly to these changes. Yet, these two knowledge systems frequently operate in isolation, continuing a history of undervaluing Indigenous perspectives [5]. A significant challenge lies in creating valuable interfaces between these systems to allow for responsible complementarity, preserving the context and utility of TEK within computationally intensive Artificial Intelligence (AI) models.

This challenge is particularly acute in the Sápmi context, where historical processes, such as the forced displacements, have fragmented the oral knowledge tradition. The Sámi author Elin Anna Labba describes this state of historical and epistemic erosion in her work on the forced relocations in Sweden and Norway:

> "This is a text based on a chorus of narrators: brief anecdotes, joiks, fragmentary memories. I have spliced and interwoven them: sometimes the colors are vibrant, sometimes the weave has holes and silence. Oral narratives have faded away over time. I have had to accept that the form of this text is necessarily like all Sámi history, like a woven vuoddaga severed by an axe. The threads have not even been torn apart; they simply come to an end, and the pattern is difficult to resume." [6]

This passage highlights a central problem addressed by this thesis. The knowledge base exists as a collection of interconnected, partial fragments rather than a complete, labelled dataset. Consequently, ethical AI systems in this domain require and architecture designed to reason with these fragments. These systems must respect informational gaps through contextual analogy instead of demanding statistical completeness. Addressing this need, this thesis explores how logical formalism and Social Practice Theory (SPT) function as a computational framework to act as an analogue for collective memory and storytelling while highlighting the limitations of modern AI paradigms.

Indigenous traditional knowledge, cultures, and languages provide a central foundation for building socio-ecological resilience in the face of rapid climate change. Consequently, there is a pressing need for a new educational paradigm in the North, one that is locally rooted yet multidisciplinary, incorporating the specific world-views and traditional knowledge of Sámi reindeer herders. To meet emerging global challenges and coordinate effectively with national and international actors, herders require the development of local competencies and essential adaptive tools. Ultimately, the future sustainability of reindeer herding societies depends on the synthesis of scientific data and traditional experience-based knowledge to educate and train the next generation of Arctic leaders [7].

This thesis responds to these calls by exploring whether Social Practice formal logics and a responsible AI approach can serve as resilient, sustainable tools for representing TEK. This exploration aims to advance the integration of TEK into scientific models and climate strategies, thereby promoting resilience in traditional Sámi reindeer herding. The goal encompasses both technical advancement and the preservation of voices and perspectives frequently marginalized in global responses. Conducted in collaboration with the TAIGA Centre for Transdisciplinary AI, utilizing support from Umeå University's Arctic research network and the Sámi research project "Sámi Trail of Tears" this work addresses the urgent need for new paradigms of knowledge integration. This thesis serves as a pilot study, establishing a foundation for the further development of Sámi AI at Umeå University.

The thesis is structured as follows: we first begin with the problem statement followed by relevant background and theory. We then detail the scientific methodology followed by results and a discussion. Finally, the thesis concludes with a conclusion and a section on future work.

2 Problem Formulation

The overarching research problem of this thesis is: **What can we learn about the richness of Traditional Ecological Knowledge TEK) by attempting to capture it in a formal representation?**

This problem arises from the recognition that traditional knowledge often transmitted through narrative, storytelling, and lived cultural practice cannot be easily reduced to large-scale datasets or abstracted into formalized structures without loss of meaning. Attempting to encode such knowledge raises fundamental questions of what is preserved, what is omitted, and how context and cultural depth are treated in the process of representation. At the same time, attempting such a translation has the potential to highlight the unique characteristics of traditional knowledge systems, thereby exposing both the limitations of science-based knowledge representation approaches and the epistemic value of diverse narrative traditions.

The problem also entails a clear ethical and legal dimension. Engaging with Sámi traditional knowledge requires adherence to general frameworks for responsible AI (e.g., EU Ethics Guidelines for Trustworthy AI, AI Act, GDPR), but also by Indigenous-specific principles. These include the *Ethical Guidelines for Research on the Sámi in Finland* [1], the Sámi Ownership and Data Access (SODA) principles [2], and the Indigenous data sovereignty frameworks (CARE principles) [3], alongside the FAIR data principle [4]. Additionally, the project acknowledges the United Nations Declaration on the Rights of Indigenous Peoples (UNDRIP)as a global rights framework. Embedding these principles ensures the framing of the problem encompasses cultural fidelity, epistemic justice, and the preservation of Sámi sovereignty over knowledge, rather than remaining solely a technical or epistemological challenge.

From this overarching challenge, three research questions are derived:

> **RQ1 Feasibility:** To what extent can traditional knowledge, particularly narrative and relational aspects, be represented within computational knowledge structures?
>
> **RQ2 Contextual Gaps:** What cultural and contextual dimensions are lost or distorted in the process of formalization, and how can these gaps be identified through comparative analysis of formal structures and original narratives?
>
> **RQ3 Narrative Potential:** How can storytelling itself provide a representational framework or background knowledge that preserves aspects of traditional knowledge which formal structures fail to capture?

[1] https://oulurepo.oulu.fi/handle/10024/50115
[2] https://www.saamicouncil.net/documentarchive/soda en
[3] www.gida global.org/care
[4] https://www.go fair.org/fair principles/

This thesis investigates the nuances of traditional knowledge through the process of formalization. It avoids reducing knowledge to rigid computational forms; instead, it investigates what the attempt to formalize reveals about richness, context, and epistemic boundaries. This inquiry contributes to broader discussions on responsible AI and knowledge representation by considering how cultural narratives and traditional epistemologies inform the design of systems that remain technically meaningful and ethically attentive.

3 Background

This chapter establishes the contextual and interdisciplinary landscape necessary for understanding the intersection of Indigenous knowledge and Artificial Intelligence (AI). It examines the role of Sámi Traditional Ecological Knowledge (TEK) in climate resilience and analyses the ethical challenges involved in integrating these perspectives with scientific models. Furthermore, it defines the limitations of statistical AI generalization when applied to the rich, context-dependent nature of traditional practices.

3.1 Sámi Traditional Ecological Knowledge

The Arctic is currently undergoing environmental transformation at a rate nearly four times the global average [8], acting as a significant destabilizer for the Indigenous populations who rely on these ecosystems for their cultural and economic survival. For the Sámi people of Sápmi, these rapid ecological shifts pose a direct threat to reindeer husbandry, a practice that is deeply intertwined with the Arctic environment.

In Western frameworks, responses to such threats are frequently framed around the concept of "stability." Indigenous communities in general and reindeer herding communities specifically, however, diverge from this notion, sustaining themselves instead through a dynamic capacity for constant adaptation [9]. The Western term "resilience" is itself rare in Indigenous vocabularies; the concept is embodied through active, relational values and cyclic relationships with the land. In Northern Sámi, this adaptive world-view can be captured by the concept of *birgejupmi*, translating roughly to "managing" or "getting by." *Birgejupmi* represents a philosophy of self-sufficiency and the ability to endure harsh conditions by taking only what is necessary, ensuring that resources remain available for future generations [10].

This adaptive capacity is encoded within Traditional Ecological Knowledge (TEK). TEK is a complex and cumulative body of knowledge, practice, and know-how, evolving through adaptive processes and transmitted across generations via cultural practice. TEK provides a foundational framework for socio-ecological resilience, offering crucial insights into subtle environmental changes that frequently remain invisible to conventional, data-driven scientific methods.

Central to Sámi TEK is the inseparable link between language and the landscape. The Sámi language functions not merely as a tool for communication but as a repository of ecological data and professional expertise. Inger Marie Gaup Eira has documented that the herders' knowledge base is encapsulated within a specialized vocabulary that is unique to their local needs [11].

This connection is most evident in the Sámi terminology regarding snow conditions, which are a prerequisite for mobility, tracking, and the availability of pasture plants [12, 11]. Eira

notes that herders in the Kautokeino area utilize approximately 318 distinct noun stems to designate various types of snow and snow metamorphosis processes. As Krupnik and Müller-Wille concur, this represents perhaps the richest snow terminology of any language in the world [13]. This lexicon is not strictly descriptive of objective physical properties; it contains semantic components referring to the utility of the snow for reindeer husbandry, the time of year, and the specific locality. Thus, the language itself constitutes a specialized knowledge base for adaptation, allowing herders to communicate subtle variations in grazing conditions that determine the herd's survival.

Elin Anna Labba powerfully illustrates this connection by describing the Sámi language as a "place" in itself; a woven structure of memory and geography. Displacing people or eroding their language, as seen in the forced relocations Labba describes, effectively "severs the pattern of knowledge with an axe." This act creates epistemic gaps that are difficult to re-establish, highlighting the fragility of oral knowledge systems when disconnected from the land and practices that sustain them [6]. This epistemic depth reflects a universal characteristic of Indigenous knowledge systems, where practices are rigorously defined by their specific cultural and ecological contexts. While Indigenous communities globally share the imperative to adapt to climate change, this thesis centres specifically on the Sámi context to explore the nuances of this adaptation.

This pattern of knowledge is further evidenced in the ethical and practical obligations that define daily life in reindeer husbandry. Qualitative accounts from Sámi practitioners highlight that Traditional Ecological Knowledge is not limited to environmental observation. Instead, it constitutes an all-encompassing framework where every action within herding is a manifestation of traditional wisdom. A central tenet of this worldview is a profound sense of duty toward the reindeer. This obligation dictates that every part of the animal must be utilized and that nothing should go to waste. Such values are physically expressed through the Northern Sámi concept *Duodji*, or traditional handicrafts. For example, the practice of *Njaldit* involves the preparation and drying of reindeer skins for use as seating or the crafting of specialized clothing. These practices demonstrate that TEK is a relational system where ecological sustainability and cultural duty are inseparable. By ensuring the total utility of the reindeer, practitioners enact a holistic philosophy that preserves both the ecosystem and the heritage of the Sámi people. This lived experience confirms that TEK remains a modern, vital force that governs the interaction between human society and the natural world.

Although TEK provides a high-resolution, historical understanding of local ecosystems, the unprecedented velocity of current climate change presents challenges that traditional heuristics alone may struggle to address. Concurrently, increasing sophistication in climate modelling and simulation offers predictive capabilities that serve as powerful assets for adaptation. Consequently, supporting Indigenous resilience requires investigating how these data-driven models can complement TEK. Historically, these two knowledge systems have operated in isolation, often resulting in the marginalization of Indigenous perspectives. Current research, however, increasingly recognizes that integrating TEK with scientific modelling generates a synergistic effect, creating value that exceeds the sum of its parts.

Future sustainability and adaptation strategies in the Arctic require a new educational and operational paradigm that incorporates multidisciplinary and holistic approaches, treating TEK and Scientific knowledge as equal. As argued by the EALÁT project, adaptation to new hazards requires the building of local competencies where reindeer herders are equipped with essential tools to meet global challenges [7]. This necessitates a co-production of

knowledge where scientific data sets and Machine Learning (ML) predictions complement, rather than replace, the experience-based knowledge of the herders.

To achieve this, we must practice de-colonial research methodologies. As Labba discusses regarding the historical displacement of the Sámi, future transitions must be implemented through de-colonizing methodologies. This implies that if AI and ML are to be used in Arctic affairs, they must be created to treat minority perspectives and Indigenous ontologies as equal to Western scientific data.

3.2 The Challenge of Integration: Ontological Friction and Epistemic Justice

Achieving United Nations Sustainable Development Goal 15 requires a fundamental shift in humanity's relationship with the natural world. While the United Nations emphasizes that addressing interconnected ecological crises demands recognizing the tremendous value of nature [14], the mechanisms proposed for this recognition often rely on Western epistemological frameworks. The co-production of knowledge, wherein scientific and Indigenous perspectives are brought into dialogue, offers a promising pathway toward a more holistic understanding of the environment. Riedlinger and Berkes [15] outline how Traditional Ecological Knowledge (TEK) enhances scientific inquiry through localized expertise, climate history, and community-based monitoring. Similarly, Rakova and Winter [16] emphasize the importance of ecosystem restoration for improving community well-being and biodiversity conservation, stressing that such efforts require the meaningful integration of TEK with data science.

However, the concept of "sustainability" is not an ontologically neutral objective; rather, it often functions within a *hegemonic knowledge system*. A hegemonic system dominates other forms of knowing by establishing its own structures, thoughts, and frames as the normative "default," thereby marginalizing alternatives. In this context, sustainability narratives frequently extend the logic of *Western Naturalism*, a paradigm that treats the world as a measurable, objective entity distinct from human agency [17]. This paradigm assumes a *closed world* wherein sufficient sensor density yields a complete picture of reality. Such perspectives frequently fail to grasp Indigenous ontologies. Within these frameworks, land functions as a relation of mutual responsibility. Sustainability narratives frequently enforce a Western world-view, co-opting Traditional Ecological Knowledge to serve institutional or technological development goals. In doing so, this approach ignores the holistic paradigms foundational to Indigenous knowledge systems globally.

3.2.1 From Physical to Digital Extractivism

The integration of these distinct knowledge systems faces severe epistemological risks, exacerbated by a history of colonialism, land dispossession, and forced assimilation. These historical injustices have fostered deep-rooted distrust [5]. This distrust is compounded by the expansion of colonial logic from the physical to the digital sphere. Originally rooted in the Latin American concept of *extractivismo*, extractivism refers to the profit-driven exploitation of natural resources. It constitutes a mode of accumulation with significant social, environmental, and political implications [18, 19]. Historically, this practice was linked to the settler-colonial exploitation of Indigenous lands for the benefit of the Global North. However, contemporary extractivist logic has transferred into new domains, including

finance, global economics, and digital infrastructure [20, 21].

Within this expanding digital sphere, data extractivism has emerged as a new mode of production. This process involves the collection, manipulation, and deployment of data, motivated by economic gain and intertwined with global financial systems [22]. Pasquinelli and Joler [23] describe contemporary Artificial Intelligence development as a "corporate regime of knowledge extractivism and epistemic colonialism." Under this regime, Machine Learning (ML) algorithms act as powerful instruments for information compression. Such compression frequently eliminates the contextual and cultural nuances inherent in Traditional Ecological Knowledge (TEK). Consequently, integrating TEK into AI becomes a highly complex political act. It risks stripping Indigenous knowledge of its relational foundations to render it compatible with capital-driven technological systems.

Crucially, this threat of digital extraction occurs alongside ongoing physical displacement. While alignments between Sámi TEK and scientific measurements demonstrate that science can enhance adaptive capacity when both systems are respected [4], reindeer herders currently navigate a landscape where climate change acts as a threat multiplier. The Sámi face a "double burden" wherein climate variability intersects with the fragmentation of land due to forestry, mining, and industrial green transitions [24]. Independent Sámi researchers [25] criticize these extraction systems for their inability to grasp Indigenous ontologies, calling instead for a framework of relational ethics. The crisis transcends a changing climate; the traditional response of moving a reindeer herd to new grounds is frequently rendered spatially unfeasible due to industrial occupation [26].

To avoid the "illusion of inclusion," research must be guided by Indigenous leadership and power-sharing relationships [27]. A responsible computational approach requires moving beyond mere inclusion to foster a space where diverse epistemologies can co-exist, ensuring that Indigenous knowledge is neither spatially erased by physical industry nor epistemologically subsumed by the extractive logic of AI.

3.3 The Epistemic Limits of Statistical Generalization Why Big Data Fails)

Artificial intelligence (AI) systems fundamentally rely on data. However, raw data alone does not constitute intelligence; for AI to reason effectively, it must transform information into structured knowledge through Knowledge Representation. Knowledge Representation provides the formats and methods by which data is organized, enabling machines to solve problems and adapt to new experiences [28].

Current dominant AI structures, particularly Large Language Models (LLMs) and deep learning architectures, rely on trend analysis and pattern recognition. These models operate by ingesting vast quantities of data to identify high-frequency patterns, predicting the most probable truth based on statistical dominance. While effective in data-rich domains, this architecture inherently prioritizes the average and actively filters out the outlier [29].

3.3.1 Statistical Smoothing as Hermeneutical Injustice

In the context of Traditional Ecological Knowledge (TEK), this statistical prioritization constitutes a systemic ethical flaw that can be viewed through the lens that Fricker [30] terms hermeneutical injustice. Hermeneutical injustice occurs when a community's social

experience remains obscured from collective understanding because the dominant structure lacks the requisite interpretive resources to make sense of it. Within algorithmic climate modelling, the dominant structure consists rigidly of Western scientific variables such as temperature, precipitation, and wind speed [31]. Consequently, when a Sámi herder articulates a condition based on relational or spiritual concepts, the AI model lacks the ontological nodes necessary to represent this input. The system effectively prevent the recognition of Sámi observation as knowledge, reducing complex socio-ecological insight to the status of statistical noise [32].

When language and culture is represented merely as data, it is inherently stripped of its situated context. This process creates a semantic gap analogous to a literal translation that captures the syntax of a foreign text while failing to convey its cultural idiom. In the case of LLMs, this issue is exacerbated because the model's context is derived from the statistical mean of a dominant, largely Western corpus [33, 34]. Even if the model encounters a specific Sámi term, the interpretive resources surrounding that term, i.e., the lived experience of the Sápmi landscape, are statistically absent. The model inevitably regresses to the mean of its training data, interpreting unique Indigenous concepts through a generalized, Western lens. Therefore, superficial alignment via fine-tuning is insufficient; while it may adjust the model's vocabulary, it cannot construct the socio-ecological embeddedness required to interpret the data correctly [31]. The model may predict the word, but it cannot compute the context.

By optimizing for generalization, statistical AI systems perpetuate a form of epistemic erasure. TEK is often specific, localized, and held by a minority population, functioning as a statistical outlier. Standard machine learning algorithms are designed to smooth out these outliers to prevent over-fitting [33]. For example, a specific Sámi observation about snow density in a particular valley is not a trend to be averaged with regional weather data. It is a discrete, critical fact that determines the survival of the herd in that specific location [4]. If an AI system averages this observation with global climate data, the specific signal is lost. Furthermore, replication across space and time is often weak in environmental sciences because phenomena are embedded in specific locations [29].

Therefore, a core challenge for the inclusion of Indigenous perspectives in AI systems is not simply adding more data, but changing the *representational framework*, the primary barrier is not a lack of data, but a lingering colonial power imbalance that renders Indigenous participation performative rather than substantive [27]. By prioritizing architectures capable of preserving narrative knowledge as discrete, inviolable logical units, the unique and context-dependent observations of Sámi herders can be maintained independently. This structural autonomy serves as a safeguard against the hermeneutical injustice mentioned above, where localized insights are often marginalized by the prioritization of high-frequency patterns. Shifting from opaque statistical correlations toward explicit logical structures establishes a robust foundation for explainability and transparency. Furthermore, this transition facilitates a form of human-AI collaboration that respects the epistemic sovereignty of traditional knowledge holders by ensuring the system's reasoning remains legible and grounded in situated meaning.

The Hegemony of Western Naturalism and Sustainability

As established in preceding sections, Western Naturalism and dominant sustainability narratives frequently operate as hegemonic frameworks. In the realm of Artificial Intelligence, this hegemony shifts from being merely ideological to computationally enforced. By demanding that all inputs conform to generalized, universally measurable variables, statistical AI systems inadvertently encode the logic of Western Naturalism and a neoliberal worldview directly into their core architectures [35].

This dynamic is exacerbated by the contemporary AI pipeline, which is trained predominantly on "WEIRD" (Western, Educated, Industrialized, Rich, and Democratic) datasets [36]. When AI models attempt to process Traditional Ecological Knowledge through these WEIRD-optimized architectures, they homogenize the information to fit pre-existing, market-driven logic. Consequently, Indigenous tales, symbols, and languages risk being "re-authored" or co-opted as mere digital assets. During this transformation, the profound relational memory and localized ecological functions of TEK are stripped away, leaving only a commodified representation of the knowledge. This process perfectly mirrors the neoliberal extractivism critiqued earlier, transposing it directly into the computational sphere [23].

To prevent AI from functioning as a tool of epistemic erasure, the field of computer science must confront a pivotal design question: is it possible to create an AI architecture that allows for the genuine co-existence of different ways of knowing? Addressing this requires acknowledging that technical problems within algorithms are frequently manifestations of underlying societal structures. Promoting AI as a tool for "sustainable development" without fundamentally altering its approach to Knowledge Representation risks reproducing colonial patterns under the guise of technological progress.

Overcoming this computational hegemony requires moving beyond opaque statistical correlations. Research on narrative intelligence and symbolic representation suggests that alternative frameworks capable of preserving causal, temporal, and relational coherence offer a vital pathway forward [37, 38]. By prioritizing architectures that treat narrative knowledge as discrete, inviolable logical units rather than data points to be averaged, developers can establish a firm foundation for the responsible design approaches detailed in the following section.

3.4 Responsible Approach to Computing Traditional Ecological Knowledge

To explore how TEK can be integrated into computational models responsibly, a hybrid reasoning framework offers a promising path. This approach combines machine learning for pattern recognition within sensor data with symbolic logic to represent norms, constraints, and community-defined priorities [39]. In this framework, forecasts from climate models can exist alongside TEK insights without erasing their epistemic origin. For instance, a weather forecast might predict a thaw event in two weeks, while a herder, drawing on environmental cues like wind patterns and snow consistency, anticipates the arrival earlier. A hybrid system allows both inputs to remain distinct yet interoperable. This supports complementarity rather than conformity and delivers usable outputs regardless of data granularity.

This approach aligns with the framework proposed by Kaur et al. [40], who advocate for the

development of *Sensible AI*, a paradigm that explicitly integrates the principles of sensemaking into algorithmic design. As the authors argue, "sensemaking subsumes interpretability." Whereas interpretability traditionally focuses on the technical transparency of a machine learning model, sensemaking expands this scope to encompass the situated, socio-cognitive understanding required for genuine human-AI collaboration. Applied to the context of Sámi reindeer husbandry, this framework demands the construction of systems where computational outputs are not merely mathematically legible, but where the lived experiences, social contexts, and epistemic authority of the herders actively drive the decision-making process.

However, the integration of these distinct knowledge systems faces systemic barriers. Eira et al. [41] emphasize that the Sámi hold unique knowledge regarding the climate and environment which has not been superseded by scientific monitoring. They argue that this knowledge must be integrated immediately into management strategies. Crucially, this exchange is only possible if traditional knowledge holders are included as equal partners and co-producers of knowledge rather than data subjects.

Achieving this equality requires navigating complex power dynamics and historical mistrust. Lauter et al. [27] caution that despite the increasing visibility of Indigenous participants in sustainability science, true integration remains obstructed by systemic power imbalances. They warn of an "illusion of inclusion" where Indigenous participation is solicited yet colonial dynamics persist beneath the surface. Western science frequently prioritizes universality and reductionism, while TEK globally operates as a relational and holistic system. Capitalizing on the synergy between Western broad contexts and Indigenous local specificity requires abandoning 'one-size-fits-all' scientific solutions. A responsible computational approach must actively resist the extraction of Indigenous data for institutional goals. Developers must foster equitable spaces where diverse epistemologies coexist, ensuring dominant logics avoid subsuming minority perspectives. Furthermore, decolonization demands a dual shift: supporting Indigenous communities in reclaiming sustainable self-determination while Western researchers relinquish claims to universal knowledge.

This challenge encompasses both political and ontological dimensions. A fundamental tension remains regarding the nature of sensing itself. Contemporary AI, driven by the era of Big Data, operates on the assumption that the world is measurable and that sufficient sensor density yields a complete picture of reality. This represents a closed-world paradigm containing only defined inputs. Conversely, humans operate in an open world. A herder senses the environment through embodied experience, engaging with subconscious cues, historical memory, and relational signs that technology cannot quantify.

This distinction highlights the "renaissance of the unmeasurable," suggesting that the most critical data for survival frequently lies beyond the reach of digital sensors. Therefore, AI agents cannot engage in true sensemaking processes in the same manner as a human agent. The machine lacks the biological and cultural substrate to experience the meaning behind the data. Bridging this gap requires a theoretical framework that embraces the loose, adaptive rules of social practice. The following chapter proposes Social Practice Theory as this necessary bridge, offering a method to formalize the context and meaning that statistical methods fundamentally omit.

4 Theoretical Framework

This chapter outlines the theoretical lenses used to investigate the complexity of Traditional Ecological Knowledge (TEK) and its potential representation in computational systems. Building on the concept of "Sensible AI" introduced in the previous chapter, we first examine Sensemaking as the fundamental cognitive process by which herders navigate uncertain environments. We then explore Social Practice Theory (SPT), arguing that it offers a computational structure capable of housing these sensemaking processes. Finally, we integrate these theories to propose that the formalization of social practices provides a step towards an "open world" system capable of reasoning with incompleteness.

4.1 Sensemaking

While traditional AI approaches often focus on interpretability, explaining how a model processes data, this thesis aligns with Kaur et al. [40] in prioritizing sensemaking. Interpretability is a property of the system (can we see the weights?), whereas sensemaking is a property of the agent-in-context (can we understand the situation?).

Sensemaking is a participatory process that involves interpreting environmental cues through social interaction and reflection to generate meaning. Through these iterative cycles, individuals and communities enact a more ordered understanding of their environment, allowing new cues to emerge and be interpreted [42]. It goes beyond perceiving the world; it is the active construction of a coherent narrative from different, often ambiguous signals. Sensemaking asserts that data does not become information until it is placed into a meaningful relationship with the actor's goals and identity.

This makes sensemaking particularly relevant for engaging with Sámi TEK, which is embedded in lived experiences and dynamic interpretations of ecological change. Although direct studies on sensemaking theory in a reindeer herding context is limited, parallel research in Arctic environments provides strong empirical support. Løland and Hällgren [43] apply this approach to study how ski guides make decisions in complex and risky terrain, showing that social context alters how ecological cues are read and acted upon. They introduce the concept of socio-ecological embeddedness, where individuals become rooted in the land by continuously interpreting spatial and temporal cues like weather, snow and topography. This method of ecological sensemaking could be a highly interesting approach, mirroring how Sámi reindeer herders navigate seasonal changes and climate variability through heuristic reasoning, environmental observation and communal knowledge-sharing.

4.2 Social Practice Theory: From Sociology to Artificial Intelligence

Social Practice Theory (SPT) offers an analytical framework that shifts focus from individual choices or broad societal rules to the "practice" itself. Instead of looking at a single person or a rigid system, SPT claims that important features of human life should be understood in terms of organized constellations of interacting persons. Thus, this theory looks at the *practice* itself through the organized activities that people do together [44]. Reckwitz [44] defines practices as routinised types of behaviour consisting of several interconnected elements: forms of bodily activities, forms of mental activities, 'things' and their use, and background knowledge.

A key point of this view is that practices are not static. They are "accepted ways of doing things" that are shared by a group and repeated over time [45]. These practices allow people to know how to act in a situation without having to make a new decision every time. For example, showering or eating dinner are recurring activities that create standard expectations for behaviour.

Originating in the foundational sociological works of Bourdieu [46] and Giddens [47], this framework argues that human life relies on routines and habits. While these early theories focused on the interplay between structure and agency, contemporary interpretations, particularly those by Shove et al. [45] and Schatzki [48], have operationalized these concepts to demonstrate how practices emerge, persist, and decay.

According to Shove et al. [45], a social practice reduces to three core, interconnected elements. This simplification is particularly valuable for this thesis as it provides a computable ontology for modelling behaviour:

> **Material:** The physical things involved, such as the human body, tools, resources, and the environment (e.g., a car, a road, or the weather).
>
> **Meaning:** The social significance, beliefs, and emotions tied to the practice (e.g., the idea that driving children to school shows "good parenting").
>
> **Competence:** The skills and knowledge needed to perform the practice (e.g., knowing how to drive safely).

Crucially, practices function as dynamic entities. They are "accepted ways of doing things" that evolve slightly with every performance, allowing for continuous adaptation [49]. Unlike models that scrutinize individual decision-making, SPT treats the practice itself as the smallest unit of analysis. This abstraction shifts the analytical focus from the specific choices of an individual to the broader structural conditions that enable the activity. This framework creates the potential for whole-system solutions rather than targeted behavioural corrections.

4.2.1 Relevance to Traditional Ecological Knowledge

Although SPT began in sociology, it serves as a powerful lens for understanding TEK. TEK functions as a living, practice-based knowledge system rather than a static repository of facts. It fits the definition of a social practice: a routine activity where knowledge is enacted through interaction with the environment.

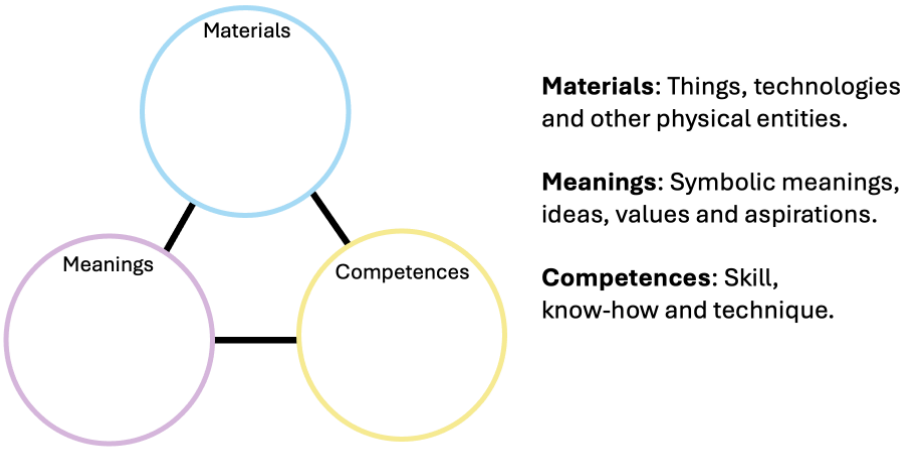


Figure 1: Shove et al.'s three-element Social Practice Framework.

In the specific context of Sámi reindeer herding, the tripartite structure of SPT maps directly to the herder's reality [4, 41]:

> **Material:** This category encompasses the biotic environment itself: the snow conditions, grazing lands, herd dynamics, and the technology used to navigate them.
>
> **Competence:** This refers to the herder's accumulated ecological literacy (TEK), such as the ability to interpret snow density or predict weather patterns. These skills function as heuristics that guide action without enforcing rigid rules.
>
> **Meaning:** The practice is sustained by deep cultural values, for example the obligation to preserve the herd and the land for future generations [41].

By employing this framework, we can analyze TEK as a dynamic system where physical conditions, ecological skills, and cultural values are inextricably linked. This approach avoids imposing external Western norms, allowing the specific norms of the herding practice to emerge from the context itself.

4.2.2 Social Practice Theory in Artificial Intelligence

SPT has been applied to Artificial Intelligence (AI) research in order to bridge the gap between rigid, pre-defined protocols and the need for flexible, context-aware deliberation in complex environments.

Specifically, this approach has been operationalized in domains such as social robotics, serious games, and human-agent collaboration to manage the unpredictability of interaction. For instance, in the field of social robotics, Dignum et al. demonstrate that social practices provide a necessary "focus of attention" for agents operating in open environments. By organizing knowledge into practices, robots can efficiently filter sensory data, looking specifically for objects or actors relevant to the current social context (e.g., looking for a projector when entering a *presentation* context), rather than attempting to process the entire environment.

Furthermore, in the domain of multi-agent planning, Miller et al. have utilized social practices to structure human-agent collaboration. The key insight from this research is that social practices function as high-level specifications that allow agents to "fill in the gaps" of a plan using their own capabilities, rather than requiring a complete, step-by-step script. This research highlights that employing social practices significantly enhances planning efficiency; it allows agents to make safe assumptions about the behaviour of others based on shared expectations, thereby reducing the computational load of deliberating every possible interaction.

Finally, in the context of serious games, Augello et al. have applied this framework to structure natural dialogues, finding that practices provide the necessary social context to drive coherent and meaningful conversations. Collectively, these applications demonstrate that integrating SPT into AI allows systems to navigate uncertainty by leveraging the "standard patterns" of human life, transforming social context from a passive backdrop into an active resource for reasoning.

Interestingly, SPT provides AI paradigm structured around the practices, norms and plans of agents rather than goal driven, answer set or rational behaviour of artificial agents. Examining contexts through SPT has allowed AI researchers to understand and contextualize emergent human behaviours such as creativity as practices. Through the abstraction of SPT the practice does not require concrete rules or descriptions to be carried out by an agent. This allows the Material, Context and Meaning to shape the emergent behaviour of creativity [50].

Beyond formalizing routine behaviour, SPT serves as a lens for understanding complex concepts like creativity. Augello et al. [51] argue that creativity is not a random spark but a structured part of daily life. By analysing creativity as a practice, they distinguish between Situational Creativity and Creativity of Habit. Adopting this lens necessitates a paradigm shift in agent architecture; rather than treating creativity as a separate module, designers can architect agents that operate within social practices while retaining the autonomy to resolve ambiguities.

This informs AI design in two ways:

1. **Flexible Structure:** Rather than relying on rigid, step-by-step scripts, AI systems utilize 'Plan Patterns.' These general guides enable the agent to employ Situational Creativity, adapting to environmental realities while adhering to social norms.
2. **Safe Improvisation:** Since improvisation occurs within the established boundaries of a practice (defined by Materials and Meanings), the agent maintains adaptivity while remaining within safety and social acceptability constraints.

For this thesis, this perspective is crucial. It suggests that applying SPT to TEK reveals how herders employ rigorous improvisation, adapting their accumulated competence to the stochastic snow conditions of each winter.

4.3 Social Practices as a Computational Framework

As stated in the previous section, in Social Sciences, social practices are defined on the basis of materials, meanings and competences. SP provide a handle for modelling the deliberation

of socially intelligent agents because they seem to combine the elements that we require for socially intelligent behaviour. This approach addresses the limitations of rigid rule-based AI systems and the opacity of data-driven models. In order to use SPT as a lens for learning about TEK this thesis adopts the formalization of SPT proposed by Dignum [52]. This framework provides a structured yet flexible logic for representing complex, situated behaviours, making it a promising candidate for capturing the nuance of TEK.

4.3.1 The "Middle Way": Balancing Structure and Autonomy

Conventional approaches to agent deliberation often fall into two extremes: fixed interaction scripts, which are too brittle for dynamic environments, or purely goal-driven deliberation (BDI), which can be computationally expensive and unpredictable. Dignum's approach positions Social Practices as a "middle way" [52].

A SP is defined through high-level specification of "normal" or expected behaviour within a given social context. It functions based on the concept of *landmarks*. Rather than specifying every micro-action an agent must take, the SP defines abstract states (landmarks) that must be reached. The agent adopts the practice based on its evaluation of the context and uses its own capabilities to "fill in the gaps" between these landmarks. This allows the system to guide behaviour via social expectations while retaining the flexibility to adapt to specific environmental conditions, a dynamic that closely mirrors the heuristic nature of TEK [11, 4].

While sensemaking can serve as a model for the cognitive process, SPT provides the structural container for that process. In the formal logic of social practices proposed by Dignum [52], a practice consists of Context, Meaning, Expectations, and Activities. The Meaning (M) quadrant of this framework is the computational home of sensemaking. In Dignum's logic, this is formalized through "Counts-as" rules (X counts as Y in Context C).

> **Input**: The "Brute Fact" (e.g., The herd is wandering restlessly).
>
> **Sensemaking Process**: The herder filters this observation through cultural values and experience.
>
> **Output**: The "Institutional Fact" (e.g., This counts as a Risk Event or Bad Guohtun).

This formalization reveals the critical limitation of purely sensor-based AI. A sensor can detect the Brute Fact (X: movement patterns), but without the Meaning structure, including the cultural values defined in (Pv) and purpose (P), it cannot derive the Institutional Fact (Y). The machine cannot "make sense" of the threat.

This synthesis of Sensemaking and SPT offers a pathway toward "Open world" AI paradigms. "Closed world" AI assumes that all relevant variables are known and defined a priori (e.g., a sensor array with fixed inputs). If a variable is not in the ontology, it does not exist for the system often assuming that better sensing equals better understanding. In contrast, the SPT formalization utilizes *Landmarks* and *Plan Patterns* rather than rigid scripts

> **Closed World Script)**: "If temp < -10, then check snow." (Fails if the temperature sensor breaks or if the wind is the actual driver).
>
> **Open World Landmark)**: "Goal: Ensure Herd Safety." The agent is free to use any available strategy or cue to reach this state.

By defining the practice rather than the input, we allow for an open-world approach where the agent (or the human-AI team) can incorporate new, undefined cues to bridge the gap between current reality and the desired landmark. This aligns with the previously argued concept that Sámi resilience is not about stability. The formal representation of TEK, therefore, is not about capturing a static snapshot of the past, but about encoding the logic of adaptation that allows herders to survive in an uncomputable, open world.

4.3.2 The Logical Architecture

To formalize these sociological concepts, Dignum propose a logic that combines **Dynamic Logic** (to model actions and state transitions) with **Epistemic Logic** (to model beliefs and common knowledge).

The formalism is built upon a language $\mathcal{L}$ that distinguishes between physical actions and social interpretations. The core logical components necessary for our analysis are:

Action Expressions) and Events ξ): The logic distinguishes between an abstract action and an event ξ performed by a specific agent or group. For example, A : denotes that a set of actors A performs action together.

Modal Operators:

- $DO\ \xi)$ and $DONE\ \xi)$: Operators indicating that an event will occur next or has just occurred.
- $Cap\ a,\ )$: Indicates that agent a has the *capability* to perform action .
- $B_a\ \phi)$ and $CB\ \ \phi)$: Represents the *belief* of agent a regarding proposition ϕ, and the *common belief* held by a group A. This is crucial for TEK, which relies on shared community knowledge and Sámi culture.
- $[\xi]\phi$: indicated that after event ξ occurs, ϕ will always be true.

This logical foundation allows us to construct a Social Practice Model composed of four distinct quadrants: *Context*, *Meaning*, *Expectations*, and *Activities*.

4.3.3 Formalizing the Components of Practice

To represent a TEK case, we must map it into the following formal structure composed of four distinct quadrants: *Context*, *Meaning*, *Expectations*, and *Activities* that expand the un structured Social Science definition of SP [52].

Context C)

The context defines the boundaries of the practice: the "where," "when," and "who." It anchors the practice in a situated reality.

Roles describe the competencies and expectations about certain types of actors. Thus a herder is expected to be able to drive the snowmobile in various weathers. We use the predicate $play\ a, r, c)$ to indicate that agent a enacts role r in context c.

Actors are all people and autonomous systems involved, that have capability to reason and (inter)act. This indicates the agents that are expected to fulfill a part in the practice.

Resources are objects that are used by the actions in the practice such as skis, snowmobile, gloves, probe etc. These resources are assumed to be available both for standard actions and for the planning within the practice.

Affordances The properties of the context that permit certain actions. This is also dependent on the match between context conditions and actor characteristics. For example, "hard snow" could afford "mobility" but could also deny "grazing." Formally, $affords\ O,\ \ ,c)$ states that a set of objects/resources O allows action to be performed in context c.

Places indicates where all objects and actors are usually located relatively to each other, in space or time: The reindeer are in the grazing pasture, the snowmobile is in the garage, the shovel is in the shed, etc.

Meaning M**)**

This quadrant bridges brute facts (physical observations) and institutional facts (social/cultural interpretations).

Purpose determines the social interpretation of actions and of certain physical situations. E.g. the purpose of making sure the reindeer can graze is survival and well being of the herd.

Promotes indicates the values that are promoted (or demoted, by promoting the opposite) by the social practice. Driving carefully promote the value of universalism, because you take care of all passenger's safety and the safety of other road users.

Counts-as are rules of the type "X counts as Y in C" linking brute facts (X) and institutional facts (Y) in the context (C). Formally:

$$M, w \models \text{counts-as}\ c, r : \quad ,\ \ _2) \tag{4.1}$$

E.g., in a voting place, filling out a ballot counts as a vote. This implies that if an actor performs physical action (e.g., observing an east wind), it counts as performing the social/epistemic action $_2$ (e.g., identifying an ice risk).

Expectations E**)**

This component replaces rigid scripts with flexible guidance.

Plan patterns describe usual patterns of actions defined by the landmarks that are expected to occur.

Norms describe the rules of (expected) behaviour within the practice. They are described as deontic rules using obligations, prohibitions and permissions.

Strategies indicate possible trigger-action combinations that can be performed at any time within the practice. Not all strategies need to be performed They denote potential courses of action, whenever the trigger is satisfied. Strategies are formalized as:

$$strategy\ \phi, DO\ B : \gamma), sp) \tag{4.2}$$

Meaning: "If condition ϕ holds, the group B is expected to do γ". This formalizes the "If-Then" heuristics (e.g., "If the valley has ice, move to the bare ridge").

Start condition, or trigger, indicating how the social practice starts. At 6:30 the sun rises and begins the social practice of reading the snow.

Duration, or *End condition*, indicating how the social practice ends.

Activities A**)**

The operational layer describing specific actions ( ) and the competences (Cap) required to execute them. In the context of TEK, competence is not just physical (e.g., digging snow) but epistemic (e.g., the ability to *interpret* snow conditions).

Possible actions describes the actions that usually occur in the social practice. This gives a kind of repository of actions with which activities in the social practice can be planned.

Requirements indicate the type of capabilities or competences that the agent is expected to have in order to perform the activities within this practice.

4.3.4 Operationalizing the Logic

In this framework, an agent engages in a cycle of sensemaking supported by the formal logic:

1. **Context Recognition:** The agent evaluates the world state w. If the Start Conditions (SC) are met ($M, w \models SC\ sp, \phi)$), the Social Practice becomes active.
2. **Landmark Navigation:** The agent looks at the Plan Pattern. It identifies the next desired state (landmark) ϕ.
3. **Gap Filling:** The agent checks its Affordances and Capabilities. It selects specific actions that will bridge the gap between the current state and the landmark ϕ, guided by Strategies and constrained by Norms.
4. **Meaning Derivation:** Through the Counts-as rules, the agent interprets physical feedback from the environment as meaningful social/ecological signals, which may trigger new strategies or norms.

This formalization provides the necessary syntax to view TEK as a dynamic, social practice of interacting with the environment.

5 Methodology

5.1 Research Design

This thesis employs a constructive design research approach, using formal logic as an illustrative tool to investigate the complexity of Traditional Ecological Knowledge (TEK). Rather than applying statistical machine learning methods, which generally require large, homogenous datasets, this research utilizes Social Practice Theory (SPT) formalized through modal and dynamic logics [52].

The research process is qualitative and abductive. It moves iteratively between narrative descriptions of Sámi reindeer herding found in secondary literature and the structural constraints of computational formalization. The core methodological objective is an analysis of epistemic limitations: systematically identifying the boundaries where standard formal logic diverges from or constrains the nuances of TEK.

5.2 Literature Selection and Source Material

As direct fieldwork was outside the scope of this study, the primary material for this research consists of authoritative secondary sources that document Sámi TEK through narrative and linguistic analysis. The selection criteria prioritized texts that provide phenomenological detail regarding the experience of herding and linguistic precision regarding Sámi terminology for snow and ice conditions.

The primary texts selected for formalization include:

> *Sámi traditional ecological knowledge as a guide to science* by Riseth et al. [4], which bridges scientific measurements of snow physics with traditional terminology.
>
> *Snö* by Ryd and Rassa [12], which offers granular, narrative descriptions of herding decisions and snow classification.
>
> *The Silent Language of Snow Sámi traditional knowledge of snow in times of climate change* by Eira [11], which provides deep analysis of snow in Sámi contexts as well as narrative research on how the Sámi language is used in traditional reindeer herding practices.
>
> *A novel siida-based monitoring system to observe effects of climate variability on winter pastures in Sámi Reindeer herding* by Eira et al., [53], which provides a novel and rich diary based monitoring system, their study synthesises and highlight various patterns and concepts of reindeer herding and its relation to topology, temperature and Sámi linguistics.

Indigenous Reindeer Herding and Adaptation to New Hazards in the Arctic by Mathiesen et al., [9], which provides context to how traditional reindeer herding practices are changing in light of climate change.

The Rocks Will Echo Our Sorrow by Labba [6], which provides historical context on displacement, framing the cultural values inherent in the social practice.

5.3 Formative Expert Consultations

To ensure the theoretical validity and ethical integrity of this interdisciplinary work, the research design included a series of formative expert consultations. These consultations were unstructured, off-the-record dialogues that have helped inform various aspects of this work. The interdisciplinary nature of the topic made these consultations important to bring dialogue and a variety of expertise to the table. The consultations involved professors, independent researchers and artists from around the world working with or within Indigenous contexts, representation in data and decision-making.

Consistent with the principles of Constructive Design Research, these interactions were not treated as primary data collection for qualitative coding. Instead, they served a distinct methodological function: to calibrate the research scope, challenge the author's prior assumptions, and refine the theoretical lenses applied to the secondary literature. To preserve the anonymity of the informants' perspectives are aggregated below by their epistemic contribution rather than by individual attribution. The insights derived from these consultations shaped the thesis in three critical dimensions:

Indigenous Methodologies and Ethical Sovereignty

Consultations with scholars in Sámi studies and members of the reindeer herding community were pivotal in framing the ethical boundaries of the project. These dialogues illuminated the risks of digital extraction, emphasizing that Indigenous data is often contextual and resistant to the commodification inherent in standard datasets. Informants highlighted the North Sámi concept of *Verdde* (a mutually beneficial exchange) as a necessary method for working with Indigenous communities. This guided the decision to adopt a non-extractive methodology and to utilize only publicly available secondary sources at this stage of the research. Anthropological experts warned against uncritically adopting "sustainability" as a metric. These experts argued sustainability itself can be viewed as a colonial approach and that Indigenous relationships with the land are fundamentally regenerative, meaning in balance, represented by the notion of *Birgejupmi* [10] and connecting back to the *Verdde* system mentioned.

Situated Cognition and Ecological Complexity

To ground the theoretical application of Sensemaking, discussions were held with practitioners in sub-Arctic ecology and researchers specializing in risk assessment in snow-covered terrain. Experts in sensemaking characterised the complexity of modelling natural cues as deceptively simple on the surface but concealing immense, interconnected complexity. This

validated the use of Social Practice Theory to capture the "messiness" of decision-making that statistical models often smooth over. Practitioners specializing in predator tracking and forest management shed light on the practical mechanics of reading biological indicators.

Critical Data and Ontological Friction

Engagement with scholars in feminist data science and critical computing highlighted the ontological friction between TEK and Western computer science. These consultations challenged the assumption that TEK can be fitted into a dataset or computational model. They argued that Indigenous knowledge was never intended for such abstraction, suggesting that any attempt to do so without new approaches would violate the ontological foundations of the knowledge itself. This insight directly motivated the use of formal logic as a critique of Big Data, rather than an attempt to replicate it.

Young Herders: Combining Tradition and Modern Technology

Discussions with active reindeer herders, particularly younger practitioners, provided a critical corrective to the reliance on historical documentation. These informants highlighted that while TEK is rigorously passed down across generations, older texts and dated documentation of herding practices can fail to capture the realities of modern herding. Young herders effectively bridge the traditional wisdom inherited from elders with a sophisticated understanding of modern adaptation strategies and technologies. This insight fundamentally shaped the analytical posture of this thesis. It served as a reminder that any formalization based on secondary literature captures a crystallized version of the knowledge, necessitating that future, applied AI systems must be co-created with the young herders who are actively navigating contemporary ecological crises.

5.4 Computational Method: The Logic of Social Practices

The theoretical lens for this work combines SPT with Sensemaking. Originating from sociology, SPT moves the unit of analysis from individual rational choice to collective practice [44]. The formal SPT framework is integrated with Sensemaking, defined as a participatory process of interpreting environmental cues [42]. The translation of TEK narratives into computational structure follows the Logic of Social Practices proposed by Dignum et al. [52]. This involves a three-step formalization process.

5.4.1 Informal Decomposition

Narrative extracts are decomposed into the four quadrants of the Dignum framework:

> **Context** (C)**:** Identifying roles (R), actors (A), affordances (Af), resources (O), places (Pl), and time (T).
>
> **Meaning** (M)**:** Identifying purpouse (P), promoted values (Pv) and "Counts-as" rules (Co).

Expectations E**):** Identifying plan patterns (PP), norms (N), strategies (S), start condition (Sc), and duration (Ec).

Activities A**):** Identifying possible actions (Ac) and competences (Re).

5.4.2 Formal Encoding

The decomposed elements are encoded using a hybrid Dynamic and Epistemic Logic. Key operators utilized include:

The Counts-as Operator: $M, w \models countsas\ c, r :\quad ,\ _2)$. This models how physical observations (e.g., East wind) count as institutional facts (e.g., Ice Risk) within the specific context of herding.

The Promotes Operator: $promotes\ sp, r :\quad , v)$. This encodes the ethical dimension, linking actions (e.g., moving the herd) to values (e.g., animal welfare).

Plan Patterns: Abstract sequences of expected actions ($\gamma\phi$) that allow for situational adaptation. This allows the model to represent the structure of a herding strategy without imposing a rigid script that would fail in unpredictable weather.

5.4.3 Epistemic Gap Analysis

The final step is the diagnostic application of the model. By attempting to run logical inferences (e.g., deriving a decision to move the herd based only on the encoded variables), the research identifies Epistemic Gaps. This methodology explicitly involves stress-testing the formal model against a "limit case"; a specific scenario where the logical derivation holds technically but diverges from the cultural or ecological reality of the situation. These points of friction highlight the tacit knowledge that ultimately resists digitization.

5.5 Researcher Reflexivity

This thesis is fundamentally shaped by my positionality as a researcher born and raised in Sweden, with an academic grounding in Interaction Technology, Design, and Computer Science. While I share the national language and certain cultural frames of reference with the majority society in Sweden, I approach the specific domain of Sámi reindeer husbandry and Traditional Ecological Knowledge (TEK) as an outsider. I am not Sámi, nor do I possess the embodied experience of living on and with the land in Sápmi. My engagement with the region is not one of lineage or livelihood, but of academic inquiry and technological design.

My disciplinary background in computer science and interaction design exerts a strong influence on how I perceive and organize information. This training instils a predisposition toward structure, logic, and optimization, a world-view that inherently seeks to categorize ambiguous phenomena into discrete variables and computable states. Engaging with the literature and the formative consultations required a conscious and frequent questioning of this *"technological solutionism"*. I have actively questioned my own assumptions and resisted the urge to smooth over the "messiness"unstructured nature of narrative knowledge just to make it fit into the logical formalism of Social Practice Theory. Instead, I learned to

embrace the friction between my computational tools and the Sámi narratives as the primary finding of the research. Although this friction initially appeared as an error requiring correction, it actually serves as a powerful signal of the ontological difference between Western computing and Indigenous knowing.

The linguistic dimension of this research also warrants reflection. While I conducted my analysis in Swedish and English, both languages in which I am fluent, the subject matter relies heavily on Sámi concepts (e.g., *guohtun*, *siida*) that possess no direct semantic equivalent in my native tongue. Although secondary sources like Ryd [12], Riseth et al. [4] and Eira [11] provide translations, I remained acutely aware that I was working with "translated worlds." The risk of semantic loss, where a complex ecological relationship is reduced to a static dictionary definition, was constant. To mitigate this, I adopted a posture of humility during the formative consultations with researchers and experts. Rather than seeking to extract data to fill my model, I used these dialogues to check the validity of my interpretations, allowing the experts to challenge the rigidity of my formal logic.

Furthermore, my position as a member of the Swedish majority culture carries historical weight. I am researching within a context shaped by colonial structures, forced displacements, and epistemic erasure, processes described vividly by Labba [6] and others. This awareness necessitated a strict ethical boundary regarding the purpose of my formalization. I was conscious of the power asymmetries inherent in digitizing Indigenous knowledge, particularly the risk of non-consensual abstraction or appropriation for external AI systems.

Reflexivity, in this context, was not merely an introspective exercise but a methodological safeguard. I maintained an analytic log throughout the formalization process, specifically documenting moments where the logic of Dignum's framework was limited in capturing the depth of the Sámi narrative. These moments of interpretive hesitation were often where my identity as a computer scientist clashed with the reality of the TEK holder. By making these clashes visible rather than neutralizing them, I aimed to position myself as a *technical translator* probing the limits of my own tools. My goal was to use the rigidity of computer science to highlight precisely what cannot be computed, thereby respecting the sovereignty of the knowledge that remains beyond the machine.

5.6 Ethical Considerations

This research aligns with the CARE Principles for Indigenous Data Governance (Collective Benefit, Authority to Control, Responsibility, and Ethics) [54] and the Sámi Ownership and Data Access (SODA) principles. The collection and integration of primary qualitative data followed strict ethical guidelines to ensure respect for Indigenous knowledge systems. Explicit informed consent was obtained from all primary informants before including their insights as empirical examples. Furthermore, all personal identifiers were removed to protect the privacy of the practitioners and guarantee complete anonymity throughout the research. The methodology prioritizes the informant's authority to control their narratives and maintains a high standard of ethical responsibility. Consequently, this approach handles Traditional Ecological Knowledge with respect and actively avoids extractive data practices.

1. **Data Sovereignty:** Primary TEK data collected during this study was used exclusively for qualitative illustration and contextual understanding. This data was strictly

excluded from ingestion into AI models or any automated processing systems. The research acknowledges the right of Sámi communities to govern how their knowledge is represented and utilized.

2. **Paradox of Participation:** The thesis explicitly addresses the risks of digitizing TEK for AI training [55]. The proposed formalization is designed as a local tool for the community. It is intended for internal use rather than as a dataset for open or global training.

3. **Non-Extraction:** The methodology is designed to highlight the limits of extraction. By demonstrating that key aspects of TEK cannot be fully formalized, the thesis argues against the indiscriminate scraping of Indigenous knowledge for global AI models.

6 Results

This project investigates what we can learn from Traditional Ecological Knowledge (TEK) through formalization. In order to analyse TEK through Social Practice Theory (SPT), we focus on a practical and narrow case study derived from Sámi reindeer herding. The case is selected for its direct relevance to Sámi resilience and its ability to demonstrate the viability and limits of SPT through iterative formalization. The structural blueprint for this case study is derived primarily from the extensive observation study, *A novel siida-based monitoring system to observe effects of climate variability on winter pastures in Sápmi Reindeer herding* by Eira et al. [53].

6.1 Case Study: The Multidimensional Assessment of *Guohtun*

The *siida*, defined as a working partnership comprising the herd, the herders, and their traditional grazing lands [11], constitutes the foundational social unit of this practice. A capacity for constant adaptation is central to the *siida*. However, this inherent dynamic capability is currently challenged by unprecedented environmental shifts. Projections indicate that Arctic winter temperatures are rising at an amplified rate, drastically shortening the snow season and reducing snow depth [8, 9]. While adaptation is deeply integrated into the herders' specialized language and practice [56], the cumulative burden of these climatic shifts alongside industrial encroachment necessitates the development of new tools. These tools must align with, rather than displace, traditional Sámi strategies.

Forage and Welfare The herders' daily work during the seven-month-long winter period focuses on identifying *ealát* (the food situation) and determining the state of *guohtun* (access to forage beneath snow cover) [56]. The concept of *guohtun* is central to the herders' conception of the reindeer's wellbeing, as it reflects the animal's need for food and water [56]. The term for herding in winter, *guođohit*, is derived from the noun *guohtun*, meaning "to let or put to graze" [56].

Protection The herder's role, known as *guođoheapmi* [56], extends to actively ensuring the herd's protection and safety. This includes preventing the herd from becoming scattered (*bieđaganan vára*), mixing with other herds (*mastanvára*), or being disturbed (*muosehuhttinvára*) [41]. The human aspect of successful herding includes making sure that the reindeer have enough food, water and shelter as well as ensuring that the reindeer can move freely while under control. This implies that the herders are aware of, and act in response to, the many needs of the reindeer and the herd. The human contribution is to take care both of the individual animal and the herd as a whole, in such a way that the needs of the reindeer are satisfied [56].

Agency Critically, the herder's task is one of constant observation and decision-making [41], often in response to the herd itself. The grazing routine is extensively dictated by the herd itself [53]. The herd's grazing behaviour is decided by, among other things, in accordance to wind direction and avoiding previously grazed areas due to the difficult *guohtun* created by a snow condition called *čiegar* (meaning the snow where the herds were grazing was very dense) [53]. This grazing routine forms part of the primary feedback loop for the herder [53]. The complexity of the herder's task lies in the holistic, language-based nature of the assessment [53]. *Guohtun* is not derived from numerical data, but from evaluating a gradual scale (e.g., "very bad *guohtun*" to "very good *guohtun*") based on simultaneous factors [53]:

Snow Physics The accessibility of forage depends fundamentally on the stratification and density of the snowpack. Hardened layers, resulting from wind compression or melt-freeze cycles, dictate whether reindeer can physically break through to the vegetation [53]. Beyond foraging, these physical properties govern the logistics of movement; the snow must be permeable enough for reindeer navigation yet sufficiently robust to bear the weight of a skier [12]. Consequently, *guohtun* is never static; it fluctuates throughout the season, requiring the herder to constantly secure optimal conditions. To navigate this dynamic landscape, herders utilize a specialized repertoire of mobility strategies [53]. These include *birravuodjit* (securing the herd by driving around its perimeter), *caggat* (actively intercepting escaping animals), *goalŋat* (guiding the herd gently to a specific zone), and *sirdit* (executing a complete relocation).

Ground Condition ***odni*)** The ground is considered the "most important factor" [53]. Types of snow that freeze solid at the bottom, such as *bodneskárta* or *bodnejiekŋa* (bottom ice), are high-risk conditions that can lead to large losses and starvation [53]. The condition of the ground before the first snow heavily influences the *guohtun* for the entire season [12]. When such conditions are dire mobility within the respective herding areas is noted as a major coping mechanism for bad *guohtun* and in response to changes in the snowpack [53].

Topography and Weather The overall assessment depends on the relationship between snow, precipitation, wind, temperature, and local topography (e.g., woodland versus plateau) [53]. Knowing and understanding how to read the herd and the environments is one of the most important skills for the herder, and is a practice also fully intertwined with the Sámi language [53].

6.2 Informal Decomposition

To apply the Social Practice (SP) framework, we construct an *illustrative scenario* analogous to a repeatable, routine human activity. This scenario isolates the daily ritual of "The Sunrise Pasture Assessment," serves as a composite illustration of a herder's morning practice dedicated to determining the current state of the environment.

The case focuses on the Herder's first inspection of the day, where multiple variables must be checked to establish the morning's "*Guohtun* Status." Crucially, this model does not

claim to be a general or empirically correct representation of all reindeer herders. Rather, by modelling a scenario derived from the literature described above, we can stress-test the limits of the SP formalization framework without extracting sensitive primary data. Future development of real-world systems must, however, be co-created with Sámi TEK holders to ensure validity.

Figure 2 details this decomposition. The final column maps the narrative elements to the formal definitions provided in Dignum's framework [52], which serve as the basis for the formal semantics presented in Figure 3. The first block of the table establishes the physical and social context. Notably, the rows for *Actors* and *Resources* pertain to specific instances (what is available *now*), whereas the rest of the definition pertains to the general practice (what is available *usually*).

The *Meaning* block links concrete physical actions to social values and purposes. The *Expectations* block is central to the framework; it outlines the sequence of events an actor anticipates, forming the basis for planning and adaptation [52]. Finally, the *Activities* block details the prerequisites actors are expected to possess and the observable actions that generate effects in the world.

Fundamentally, social practices form a structure that arises from the bottom up. Instead of forcing compliance through rigid rules, interaction patterns indicate *expectations* regarding an actor's behaviour. Consequently, there is no guarantee that behaviour will occur exactly as expected. This distinction significantly influences the formal specification, which focuses on possibilities and priorities rather than prescriptive scripts.

Within this structure, *Expectations* are tied to *Roles*, which determine who is supposed to take initiative at specific junctures. *Norms* delineate the boundaries of acceptable behaviour (permissions and prohibitions), while *Plan Patterns* provide the temporal ordering of phases, each with a specific start and end condition. Thus, the practice is fulfilled not by executing a code, but by following a trace consistent with the Plan Pattern. With this concrete description established, we proceed to the formal representation of the case study.

6.2.1 Decomposition via Shove et al.'s Framework

Before applying logical formalism, the sunrise pasture assessment case is decomposed into its constituent elements: Materials, Competences, and Meanings.

Materials The "Stuff"): The practice relies on biotic and physical resources including the herd, the snowpack (with specific states such as *oppas* (untouched snow), *čiegar* (hard snow), and *bodneskárta* (ice near ground)) , and the vegetation (*ealát*). It also includes technologies such as the snowmobile (implied by the "driving circle").

Competences The "Skill"):

- *Epistemic Competence:* The ability to "read" the herd's behavior as a proxy for environmental conditions (e.g., distinguishing if the herd is calm or wandering).
- *Physical Competence:* The ability to verify *bodni* (ground) conditions by digging and assessing the "snow physics" to determine if reindeer can access vegetation.
- *Navigational Competence:* Knowledge of alternative grazing grounds and wind-based herd movement patterns (e.g., against the wind in autumn).

Meanings The "Why"):

- *Animal Welfare:* Ensuring the herd has *guohtun* (access to food) and protection from predators.
- *Control:* Preventing the herd from scattering (*bieđagananvára*) or mixing with other *siidas*.
- *Resilience:* The aspiration to adapt to changing weather events (e.g., icing) to ensure survival.

6.2.2 Decomposition via Dignum's Framework

Figure 2 details the "Sunrise Pasture Assessment" and serves as the decomposition phase of the Social Practice (SP) formalization, incorporating the rich practice of Sámi reindeer herding directly into the four structured quadrants of the theoretical framework. This process is valuable as it promotes reflection and allows for critical input from stakeholders without fluency in logics. This can enhance alignment and transparency in the process.

The initial effort focused on accurately situating the practice within the *Context* and defining the participants. This required a conceptual shift: treating the Herd not as a passive resource, but as an active *Role* in the social practice. This honours the agency the herd itself possesses, defining its own routines for grazing and acting as a primary source of environmental feedback for the herder [53]. Furthermore, the *Activities* quadrant was detailed to capture both physical actions, such as driving around the perimeter of the herd (*birravuodjit*), and the essential non-physical task of competence the ability to read and differentiate highly specific snow conditions.

The *Meaning* quadrant proved essential for capturing the interpretive dimension of the knowledge. It is here that the *Counts-as* rules transform objective phenomena into necessary institutional facts. For example, determining *guohtun* (forage accessibility) is not based on single measurements, rather it is a composite observation requiring the conjunction of physical evidence (e.g., confirming the top ice layer is thin enough to break) and behavioural evidence (e.g., observing the herd is calm and collected) and extended by cultural or traditional beliefs. This complex, layered interpretation is what the formalization aims to preserve. Similarly, the *Expectations* section captures the necessary flexibility of the herder's cognitive plan, outlining necessary *Norms* (e.g., prohibitions against mixing herds) and dynamic *Strategies* (heuristics) that govern adaptive actions based on continuous feedback.

The completion of this decomposition establishes the necessary groundwork for the subsequent logical formalization. The next stage of this research involves translating the informal relationships defined in the table (e.g., the *Counts-as* rules and dynamic *Strategies*) into the explicit syntax of the modal and dynamic logic operators introduced in Section 4.3.

Figure 2: Informal Model of the Social Practice: The Sunrise Pasture Assessment

The Sunrise Pasture Assessment		
Context C): The Situated Reality		
Roles	**R**	**Herder**: The decision-making agent. **Herd**: An active agent that dictates its own routine and behaviour patterns [53]
Actors		(Instance Specific: e.g., A specific herder and their Siida's herd)
Affordances	**f**	**Snow Condition**: Affords (allows) or inhibits digging based on hardness [53]. Affects mobility of the herd and herder **Ground**: Ice on ground affords or denies access to forage.
Resources	**O**	Snowmobile, Grazing Pasture, The Herd (as asset), Ski pole, Shovel, etc.[12, 53]
Places	**Pl**	Winter Pasture areas (e.g., Tundra or Woodland) [53].
Time	**T**	Winter Season (approx. 7 months). Daily (at sunrise) [53].
Meaning M): The Interpretive Layer		
Purpose	**P**	To evaluate *guohtun* (forage access) to ensure herd survival [53].
Promoted Values	**Pv**	Survival of the herd. Welfare (keeping the herd together and under control) [9].
Counts-as	**Co**	**Physical Fact + Behavioural Fact** **Institutional Fact**: *Observation*: Herd is quiet. *Counts as:* positive *guohtun*. *Observation*: The bottom of the snowpack is good *Counts as:* The reindeer can reach the vegetation underneath [53].
Expectations E): Guiding Behaviour		
Plan Pattern	**PP**	*Start* *Birravuodjin* (Monitor Perimeter) *Physical Sensing* *Evaluate Status* *Decision* (Stay or Move) *End* [53].
Norms	**N**	Obligation: Check the ground thoroughly. Prohibition: Do not scatter or mix the herd with neighbours.
Strategies	**S**	**Heuristic 1**: If *bodni* is icy, move the herd to a fresh area. **Heuristic 2**: If Herd is restless, investigate hidden variables (e.g., bad smell, predators) [53].
Start Condition	**SC**	Herder begins the morning shift/initial pasture inspection.
Duration	**EC**	Until the Herd is settled and Grazing (safe status achieved).
Activities A): Observable Actions		
Possible Actions	**c**	Herders use several herding methods in winter, for example they drive around the herd (*birravuodjit*), bring it very gently to a certain area (*goalŋat*), stop the herd from escaping (caggat), or move the herd (*sirdit*) [53].
Competences	**Re**	**Epistemic Skill**: Ability to immediately characterize *Guohtun* using multiple factors simultaneously [53]. **Tactile Skill**: The ability to feel and differentiate snow layers and density [12].

6.3 Formal Encoding

This section operationalizes the informal decomposition in the previous section to formally encode the qualitative data regarding winter herding strategies. Specifically the coupling of reindeer, snow, and area described by Eira and Mathiesen [53] in Table 6 of their paper and the descriptions from Ryd and Rassa [12], especially on winter grazing in chapter 13. Based on the insights provided by the source text, we can start developing a formalization of the deconstructed case presented in figure 2. By integrating these sources, we can expand the formalization of the "Sunrise Pasture Assessment" to include the specific phenomenological checks used to verify *guohtun*.

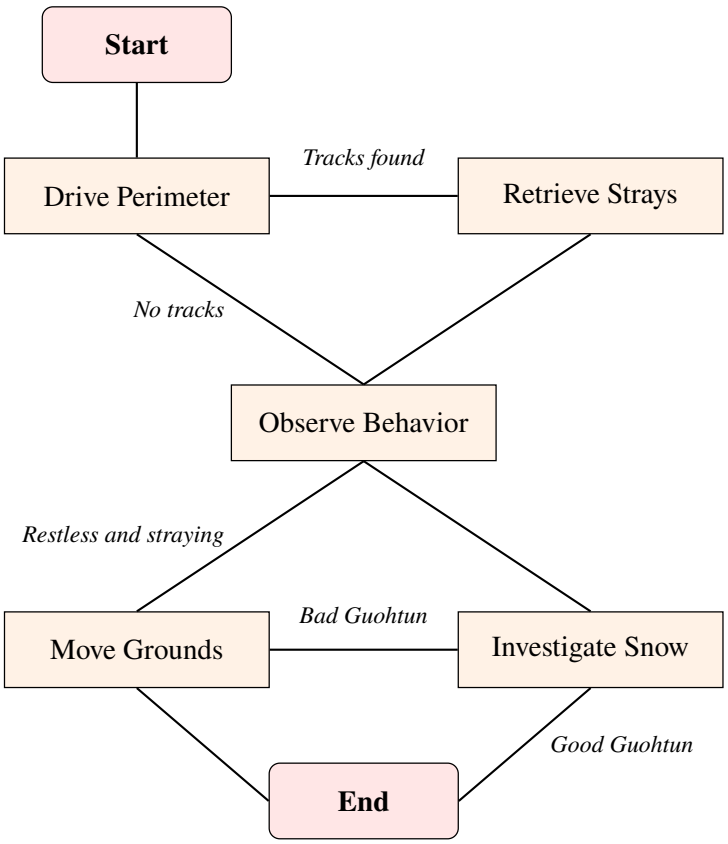


Figure 3: Flowchart of Winter Herding Plan Pattern

Using the syntax $\mathcal{L}$ defined in the theoretical framework, we encode the herder's heuristics derived from the source text.

6.3.1 Context C)

Practice: sp_{sr} = Sunrise Pasture Assessment.

Roles R**):**

- *Herder:* The observer and decision-maker.
- *Herd (r_{herd}):* An active agent possessing agency and memory. The herd is the primary agent of feedback; their behaviour dictates the assessment. As

documented by Ryd [12], the herd "decides" grazing speed and routes based on perceptions and the environment: "Reindeer are much calmer when digging is easy... if one happens to scare them, they stay put" [12]. Furthermore, in spring, "the reindeer migrate west voluntarily and find their way... they follow the same route west as they moved east in autumn" [12].

Resources O):

- The digging staff/ski pole: Used to probe consistency and ground ice without fully digging manually [12]. Similarly the reindeer hoofs act like a digging resource for the reindeer [12].
- *Suoves* (The digging crater): A resource created by the reindeer that serves as an indicator for the herder.

Affordances $\mathcal{A}f$): The physical properties of the snow determine the possibility of specific grazing actions. Ryd distinguishes snow conditions based on the kinetic energy required to access food:

$$\begin{aligned} &affords\ snow_{litna}, digging, sp_{sr})\wedge \\ &affords\ snow_{tsabma}, striking, sp_{sr})\wedge \\ &affords\ snow_{tjuohke}, \neg grazing_{ground} \wedge grazing_{arboreal}, sp_{sr}) \end{aligned} \tag{6.1}$$

This formalizes the distinction that soft snow ($litnaguohtom$) affords simple digging ($goajvvot$), hard crust ($tsabma$) requires striking with hooves ($tsabmat$), and ground ice ($tjuohke$) denies ground access entirely but affords eating lichen from trees [12].

Herd Autonomy When the herd initiates movement based on instinct, impressions or memory (e.g., seasonal migration), the herder's role shifts from director to follower/monitor.

$$strategy\ herd_initiates_migration, DO\ r_{herder} : follow_and_monitor), sp_{sr}) \tag{6.2}$$

This formalizes the observations such as "in spring the reindeer migrate west voluntarily" [12]. The herder then reacts to the herds agency and follows the impressions and decisions made by the herd.

6.3.2 Meaning M): The "Counts-As" Rules Co)

The core of TEK lies in interpreting brute facts as institutional (herding) facts. The provided text affords the following formalizations:

Rule 1: Interpreting Herd Stability

> "When the herd stays in one place, this means that the reindeer are finding food." [53]

$$M, w \models \text{counts-as}\ sp_{sr}, \text{herd_stationary}, \text{good_guohtun}) \tag{6.3}$$

Rule 2: Interpreting Herd Instability

> "When the herd starts to go from the area... check why... is it caused by predators or poor grazing conditions?" [53]

$$M, w \models \text{counts-as } sp_{sr}, \text{herd_wandering}, \text{risk_event}) \tag{6.4}$$

Here, the physical observation of "wandering" counts as a "risk event" which triggers a specific investigative strategy.

Rule 3: Wind Direction Indicators

> "In autumn: the reindeer are wandering against the wind... If the wind turns around, the herd will turn also." [53]

$$M, w \models \text{counts-as } sp_{sr}, \text{wind_shift}, \text{herd_direction_change}) \tag{6.5}$$

Rule 4: The Semiotics of Digging Debris The herder interprets the color of the snow thrown up from the crater to determine the condition of the ground (*bodni*).

> *"Det syns på den uppgrävda snön vid suoves... om betet är bättre eller sämre. Betet är bra om snön som renen sparkar upp ur gropen är vit och utan skräp."*
> ("It is visible on the snow dug up at suoves... whether the pasture is better or worse. The pasture is good if the snow the reindeer kicks up from the pit is white and without debris.") [12]

$$M, w \models \text{counts-as } sp_{sr}, \text{white_throw}, \text{loose_bodni}) \tag{6.6}$$

$$M, w \models \text{counts-as } sp_{sr}, \text{debris_throw}, \text{hard_bodni} \vee \text{ice_risk}) \tag{6.7}$$

Rule 5: The Geometry of the Crater The shape of the digging crater functions as a proxy for the reindeer's energy expenditure and stress levels.

> *"En bra grop är rund, det är suoves, en dålig grop är avlång, rågge... Gropen blir avlång som en kanal när det är tungt och svårt att gräva."*
> ("A good pit is round, it is suoves, a bad pit is oblong, rågge... The pit becomes oblong like a canal when it is heavy and difficult to dig.") [12]

$$M, w \models \text{counts-as } sp_{sr}, \text{shape_round}, \text{good_guohtun}) \tag{6.8}$$

$$M, w \models \text{counts-as } sp_{sr}, \text{shape_oblong}, \text{stress_event} \wedge \text{bad_guohtun}) \tag{6.9}$$

Rule 6: Topographical Logic Specific terrains are interpreted based on how they facilitate digging mechanics.

> *"I branta backar ligger snön oftast tunnare... snön rasar undan av sig själv från gropen."*
> ("In steep slopes, the snow is usually thinner... the snow slides away from the pit by itself.") [12]

$$M, w \models \text{counts-as } sp_{sr}, \text{slope_terrain}, \text{energy_efficient_digging}) \tag{6.10}$$

Formalism vs. Probabilistic Learning It is crucial to distinguish these *Counts-as* rules from the output of statistical Machine Learning (ML) models. In an ML paradigm, an observation of white snow might correlate with loose ground with an 85% probability. In the logic of Social Practices, however, the *counts-as* operator establishes an *institutional fact*, a constructed "truth" within the practice that serves as a solid foundation for action. This rigidity is both a strength and a weakness: it provides clarity and shared understanding (institutional stability), but risks losing the nuance of borderline cases that a probabilistic model might capture (albeit without understanding).

6.3.3 Expectations E): Strategies, Plan Patterns, and Norms

Strategies represent the conditional logic of the herder, the "If-Then" heuristics derived from generations of reading nature.

Strategy 1: The Perimeter Check

> "The herder drives around all the tracks... to find out if some animals have strayed... If so he has to get them back." [53]

$$strategy \text{ tracks_crossing_circle}, DO\ r_{herder} : \text{retrieve_strays}), sp_{sr}) \tag{6.11}$$

Strategy 2: The Grazing Investigation If the herd is wandering, the herder checks the snow physics, including the bottom (*bodni*) and vegetation (*ealát*).

$$strategy \text{ herd_wandering}, DO\ r_{herder} : \text{dig_snow; assess_bodni}), sp_{sr}) \tag{6.12}$$

Strategy 3: Relocation Logic

> "If the guohtun deteriorates, the herder needs to look for oppas... and move the herd." [53]

$$strategy \text{ poor_guohtun}, DO\ r_{herder} : \text{scout_oppas; move_herd}), sp_{sr}) \tag{6.13}$$

Strategy 4: The "Waiting" Heuristic Tsabma) When a hard crust forms on top, the strategy changes based on the time of day and solar radiation.

> *"Under vårnätter är skaren hård, men på dagen blir det varmt så renarna inväntar att snön mjuknar och behöver inte slå."*
> ("During spring nights the crust is hard, but during the day it gets warm so the reindeer wait for the snow to soften and do not need to strike.") [12]

$$strategy \text{ crust_top} \wedge \text{daytime_sun}, DO\ r_{herd} : \text{wait_for_softening}), sp_{sr}) \tag{6.14}$$

Strategy 5: The Arboreal Shift Tjuohke) When ground grazing is locked by ice (*tjuohke*), the herder must shift the herd's focus to trees.

> *"När det är tjuohke kan de ändå inte äta någonting på marken... Den enda mat renarna kan få... är laven som finns i träden."*
> ("When there is tjuohke, they still cannot eat anything on the ground... The only food the reindeer can get... is the lichen found in the trees.") [12]

$$strategy \text{ ground_locked}, DO\ r_{herder} : \text{move_to_forest}), sp_{sr}) \tag{6.15}$$

Strategy 6: Manual Intervention Bállet) If the herd is passive due to hard snow, the herder actively intervenes to "start" the grazing behavior.

> *"Skottandet är till för att få igång renarna, de ska upptäcka att det är bra bete i backen."*
> ("Shoveling is to get the reindeer started, they must discover that there is good pasture in the ground.") [12]

$$strategy \text{ good_ground} \wedge \text{passive_herd}, DO\ r_{herder} : \text{manual_dig}), sp_{sr}) \tag{6.16}$$

Plan Patterns PP)

We synthesize these elements into a Plan Pattern (γ_{daily}), a sequence of expected "landmarks" (states to be achieved). This allows for the "situational creativity" inherent in TEK. In contrast to a rigid script plan patterns allow the holistic social practice including situations and observations to affect the path in completing the landmarks.

1. **Landmark 1: Control Established $\phi_{control}$)**

   Action: r_{herder} performs `drive_perimeter`.

   Check: Are there tracks leaving the circle? If yes, execute *Strategy 1*.

2. **Landmark 2: State Assessed $\phi_{assessed}$)**

   Action: r_{herder} observes herd behavior (calm vs. agitated).

   Meaning derivation: Apply *Counts-as Rule 1* or *Rule 2*.

3. **Landmark 3: Grazing Verified $\phi_{verified}$)**

   Condition: If $\phi_{assessed}$ `risk_event`:

   Action: r_{herder} performs `dig_snow` to check for *bodneskárta* (ice) or *čiegar* (hard snow).

4. **Landmark 4: Safety Secured ϕ_{safe})**

   Condition: If `poor_guohtun` is confirmed:

   Action: r_{herder} initiates `move_herd` to `alternative_grounds`.

Plan Patterns as Sensemaking Cycles Rather than viewing Plan Patterns as mechanical sequences of actions, they should be understood through the lens of sensemaking, as a *Sensemaking Cycles*. The "Strategies" defined above are not rigid algorithms but heuristics used to navigate ambiguity and generate meaning from new data. For instance, the "Perimeter Check" is a strategy to make sense of the herd's stability before any physical intervention occurs.

6.3.4 Activities A): Competence and Verification

The formalization of activities distinguishes between the physical act and the epistemic competence required to perform it effectively.

Epistemic Competence Cap**):** The ability to use the ski pole (*goajvvomsoabbe*) effectively requires distinguishing tactile feedback.

> *"Vi var så vana att känna och avgöra vad det var för sorts snö... Med stavändan kan man snabbt känna om botten är bra... Den sjunker lätt genom lössnön, men så blir det tvärt motstånd."*
> ("We were so used to feeling and determining what kind of snow it was... With the staff end, one can quickly feel if the bottom is good... It sinks easily through loose snow, but then there is abrupt resistance.") [12]

$$Cap\ herder, \text{differentiate_tactile_resistance}) \tag{6.17}$$

Physical Actions):

- $_{check}$: Probing snow depth and consistency with a staff.
- $_{dig}$: Creating a test pit (*bállet*) to visually inspect lichen availability.
- $_{read}$: Observing the geometry of herd digging craters (*suoves* vs *rågge*).

6.4 Epistemic Gap Analysis: The Limits of Formal Logic

The formalization presented in Sections 6.1–6.3 demonstrates that Social Practice Theory (SPT) can successfully capture the structural dynamics of Sámi reindeer herding; mapping roles, routines, and conditional heuristics. However, the core methodological objective of this thesis is not merely to build a working model, but to delineate its structural limitations (see Section 5.1). By stressing the logical architecture against the nuanced narratives found in the source literature, we can identify specific Epistemic Gaps, points where the computational representation encounters structural boundaries because it cannot fully encapsulate the richness of TEK.

This section presents specific "limit cases" where the formal logic diverges from the herder's reality, illustrating the risks of reductionism.

6.4.1 Gap I: The Ontological Blind Spot The "Invisible" Variable)

The primary mechanism of the formal model is the counts-as operator, which translates physical observations into institutional facts (e.g., Equation 6.5: "White snow thrown from a crater counts as loose ground"). In the logic of the model, this is a sufficient condition for deriving $good_guohtun$ (good grazing).

The Limit Case: Consider a scenario where the physical inputs satisfy the formalized rule: the snow is white, the crater is round, and the depth is manageable. The logic derives:

$$M, w \models good_guohtun \qquad Strategy\ Stay) \tag{6.18}$$

However, a herder standing in that same spot might reject this conclusion and decide to move the herd immediately. This divergence occurs because the standard formal ontology is closed, whereas the herder's ontology is open. The herder may detect a variable that was not encoded in the "Resources" list, such as the faint smell of mould at the base of the snowpack or a "hollow" sound when striking the snow [4].

In a standard rigid system operating on a "Closed World Assumption," this generates a False Positive. The system assumes that only defined variables exist; therefore, if the known variables are safe, the pasture is safe. It lacks the "interpretive resources" (to use Fricker's term [30]) to process the undefined olfactory or auditory data.

The Social Practice formalization proposed here addresses this by introducing *dynamic semantics*. Unlike rigid models where $X + Y = Good$, the semantics of Social Practice Theory function as $X + Y = Possibly\ Good$. This shift from absolute to conditional sufficiency creates a "localized open world" model. It asserts that while the conditions for a good pasture appear to be met based on available data, the conclusion remains tentative. This logical structure explicitly reserves space for the "unknowns"; flexibility, uncertainty, and the localized nuances that only a human agent can verify. Thus, the formalization allows the system to remain "open" to undefined sensory inputs, treating the computed output as a heuristic for inspection rather than a directive for action.

6.4.2 Gap II: Agent Granularity The Herd is Not a Monolith)

In Section 6.3.1, the herd is formalized as a single collective agent (r_{herd}). The Affordances (Equation 6.1) dictate that if the snow hardness ($tsabma$) exceeds a certain threshold, the action $digging$ is denied.

The Limit Case: The model detects a hard crust ($tsabma$). The logic evaluates the affordance:

$$affords\ snow_{tsabma}, digging, r_{herd}) = False \qquad Strategy\ Move) \tag{6.19}$$

The system advises the herder to relocate to conserve the herd's energy. In reality, a herder might observe the same crust and decide to stay. This decision relies on a granular understanding of the herd's internal social structure, which the simplified agent model (r_{herd}) obscures. The herder knows that the large bulls (*sarvves*) possess the strength to break through the crust, thereby opening grazing pits for the weaker calves and females [56].

The formal model's primary limitation here is one of Agent Granularity. By averaging the herd into a single "Role" with average capabilities, the model creates a statistical generalization that erases the crucial interdependence between specific animals. This mirrors the "Hermeneutical Injustice" described in Section 3.3.1, the logic smooths out the specific "outlier" (the strong bull) that makes the practice viable, leading to an incorrect and potentially energy-expensive recommendation to move.

Finally, the formalization of the herd as a single role (r_{herd}) represents a grave simplification. As Ryd describes, snow conditions affect animals differently based on their physical capacity:

> "När det är sämre för renen blir också gropen sämre... Markbetet kan vara fint... Men ovanpå börjar vara både för hård och för tock. Renarna orkar knappt gräva gropar så de håller sig levande. Större renar kanske orkar gräva igenom och få

> till en grop, men inte mindre."
> ("When it is worse for the reindeer, the pit becomes worse... Ground pasture might be fine... but the top is too hard. Larger reindeer might manage to dig through, but not smaller ones") [12].

By treating the herd into a single agent, the model misses critical threshold effects where a "safe" condition for the bull is a "fatal" condition for the calf. This highlights the epistemic limit of statistical generalization discussed in Section 3.3.

6.4.3 Gap III: The Severed Pattern Historical and Temporal Context)

Finally, the logic operates primarily in the present tense, evaluating current state w to determine the next action. However, as Labba describes, Sámi knowledge is a "woven pattern" stretching across time [6].

The Limit Case: Consider a scenario where the formal model calculates that a specific valley offers optimal *guohtun* (grazing) based on physics: snow depth is low, biomass is high, and predator risk is zero.

$$M, w \models optimal_grazing \qquad Strategy\ Move_to_Valley) \tag{6.20}$$

Logically, the decision is "safe." Yet, the herder avoids the area. This decision may be rooted in institutional memory that is invisible to the sensor data, for example, the knowledge that this area is a traditional calving ground reserved for May, or a site associated with historical trauma or bad luck. The formal logic optimizes for immediate utility (e.g., energy), whereas the social practice optimizes for long-term continuity and cultural relations. The current logics framework is structurally unable to process the "more-than-human" experiences and sensations as a variable because it operates on a physicalist ontology. The model is logically valid but culturally wrong, demonstrating the inability of Western computational structures to fundamentally grasp Indigenous ontologies where land is not merely a resource but a relation.

A fundamental limitation is that AI agents cannot "sense" or gather information in the embodied manner of a human herder. As Eira notes, herding involves a "silent language" of tactile and sensory feedback [11]. Furthermore, AI cannot experience the "more-than-human" dimensions of the practice, such as receiving guidance through dreams or sensing the "spirit" of the trees. These are factors that can be decisive in traditional decision-making. These ontological categories do not exist in the vector space of an AI, leading to a "hermeneutical injustice" where vital knowledge is filtered out as noise [30].

Conclusion of Gap Analysis

These structural limitations do not render the formalization useless; rather, they clarify its role. The formal logic successfully captures the syntax of herding (the rules and routines) but encounters hard boundaries when attempting to encapsulate the full semantics (the lived meaning and sensory fullness). This confirms that any AI system designed for Sápmi cannot be autonomous; it must function as a decision-support tool where the human herder remains the ultimate arbiter, bridging the epistemic gaps that the logic cannot span.

7 Discussion

This thesis set out to investigate the epistemological and technical limits of Artificial Intelligence (AI) when applied to the rich, context-dependent domain of Sámi Traditional Ecological Knowledge (TEK). Motivated by the escalating threats to socio-ecological endurance in Sápmi [4] and the risks of "hermeneutical injustice" posed by statistical AI models [30], this research employed a constructive design approach. By attempting to encode the narrative practice of the "Sunrise Pasture Assessment" into the formal logic of Social Practice Theory (SPT) [52], we performed a boundary analysis: using the rigidity of code to highlight precisely what resists digitization.

Before directly answering the research questions, it is critical to situate these findings within the broader paradigms of Computer Science, data architecture, and Indigenous sovereignty.

7.1 Strategic Abstraction

While this thesis focuses specifically on Sámi TEK, the boundary analysis conducted here offers a highly constructive methodology for broader Computer Science paradigms. Digitization bias can be a loss of fidelity when translating the real world into code. However, this research demonstrates that when applied deliberately, this digitization can be leveraged as *strategic abstraction*.

This concept finds a direct parallel in robotics mapping paradigms. In robotic navigation, a system does not always require a millimetre-perfect, high-resolution metric map of every leaf in a forest to navigate successfully. Often, a topological map, which defines key nodes (landmarks) and the connections between them while abstracting the exact physical geometry, is far more robust in dynamic environments.

Applying this topological approach to Social Practice Theory allows us to utilize a "variable resolution" of digitization. In modelling the "Sunrise Pasture Assessment," we explicitly zoom in with high precision on the elements that dictate the structure and safety of the practice: the *Norms* (e.g., the strict prohibition against mixing herds) and the *Landmarks* (e.g., verifying that the herd is secure). Conversely, we deliberately "zoom out" and leave the execution details blank. The formal logic does not need to compute *how* the herder reads the tracks in the snow, nor the specific embodied mechanics of *how* they collect a stray reindeer. Those tacit, embodied skills are intentionally abstracted.

Crucially, for this variable resolution to succeed, the AI architecture must be transparent about its own boundaries; it must be explicitly clear to the user when the system has abstracted a process, why and by whom.

7.2 Architecting Data Sovereignty

Addressing these computational limitations requires rethinking the physical and systemic architecture of AI. True data sovereignty for the Sámi cannot be achieved through only ensuring that datasets are "small" or kept out of global Large Language Models (LLMs). The entire pipeline from data collection to algorithmic processing and hardware infrastructure must respect epistemic justice and data ownership.

Conceptually, this requires a paradigm shift toward free-standing, localized computational ecosystems. Rather than relying on cloud-based architectures owned by multinational corporations, Sámi communities must have the capacity to process their own data internally. A sovereign architecture would allow a community to run localized models on private infrastructure, inputting their specific knowledge and observations to generate predictions, models or coordinate strategies. Critically, the insights generated by this system would remain completely internal, free to be utilized exclusively for the benefit and adaptation of the community. Data sovereignty, therefore, is not just the right to withhold data, but the infrastructural power to compute it independently.

7.3 Indigenous-Led AI: Beyond the Academic Case Study

Finally, if new computational logics and sovereign architectures are to be developed, they must be driven by Indigenous leadership, questions, and issues. A fundamental danger in de-colonial computing is the tendency for Western researchers to use Indigenous knowledge merely as an "interesting use case" to test novel computer science frameworks.

If an AI system is not actively anchored in solving the real, self-identified issues facing the community such as mitigating the impacts of industrial land encroachment or adapting to rapid climate shifts it remains an extractive exercise, regardless of how ethically the data was handled. To move beyond the "illusion of inclusion," Indigenous AI must be conceptualized, governed, and deployed by the Sámi, for the Sámi; providing adaptable, sovereign tools that serve the priorities established by traditional knowledge holders.

7.4 Answering the Research Questions

RQ1: Feasibility of Formalization

To what extent can traditional knowledge, particularly narrative and relational aspects, be represented within computational knowledge structures? 2

The research demonstrates that formalizing TEK is structurally feasible but semantically partial. Using Dignum's framework, the study successfully mapped the syntax of herding: the roles (R), resources (O), and the conditional heuristics ($Strategies$). The concept of "Plan Patterns" proved effective in capturing the flexible routines of the herder, representing improvisation as an adaptation under rules rather than rigid adherence to a script.

However, a critical distinction emerged between the coordination of a practice and its enactment. While the logic successfully represents that a decision is made (e.g., the necessity of assessing *guohtun*), it fails to encode the embodied competence required to execute it.

This limitation, however, supports the utility of Social Practice Theory in this domain. SPT allows for a level of abstraction where the AI system manages the structural expectations (the "Plan Pattern") while deliberately leaving the specific execution details, the how, open to human agency. Consequently, the formalization functions as a navigational aid that coordinates high-level goals without attempting to simulate the irreducible complexity of the herder's embodied skill.

RQ2: Contextual Gaps

*What cultural and contextual dimensions are lost or distorted in the process of formalization?*2

Through the Epistemic Gap Analysis, three specific dimensions of loss were identified:

1. **Ontological Blindness:** The logic operates on a "Closed World Assumption," leading to false positives when critical sensory variables (such as the smell of mould or the hollow sound of ice) are not explicitly predefined in the code. In the language of SPT, this represents an limitation in the ability to digitize the element of Materials. The formal logic treats snow as a static variable, whereas the herder interacts with it as a dynamic, resisting force.
2. **Agent Granularity:** Formalizing the herd as a single agent (r_{herd}) erases the internal social dynamics of the animals (e.g., the role of lead bulls breaking the crust). This reductionism risks generating energy-inefficient advice that contradicts local realities [56].
3. **Temporal Severance:** As described by Labba [6], Sámi knowledge is a "woven pattern" of historical continuity. The formal model, optimizing for immediate utility in the present state (w), failed to account for "invisible" constraints such as historical trauma or future calving needs, thereby severing the thread of institutional memory. This highlights the difficulty of encoding the element of Meaning. The logic captures the functional goal (feeding the herd) but omits the purposeful structure; the historical trauma and cultural obligations that implicitly guides the herder's prioritization.

RQ3: Narrative Potential

*How can storytelling itself provide a representational framework?*2

The research concludes that narrative cannot merely be "input data" for an AI; it must function as the **interpretive wrapper** for the system. The formal logic acts as a skeleton, providing structure and consistency, but the narrative provides the the interpretive resources required to make sense of the logic. Consequently, AI systems in this domain must be designed not as oracles that output answers, but as storytelling partners that surface patterns for human interpretation [57].

7.5 Limitations of the Study

While this thesis provides a foundational logical architecture, several limitations characterize the research design. First, the formalization relies exclusively on secondary sources and

textual accounts of TEK. Although these sources are authoritative, they represent "crystallized" knowledge rather than the fluid, real time decision making of active herding. Consequently, the constructed models may reflect an idealized version of the practice that lacks the dynamics of a live environment.

Second, the *Limitation Analysis* is theoretical. The "Epistemic Gaps" were identified through logical comparison rather than user testing. Without deployment in a live *siida*, we cannot verify if the identified gaps are the only gaps, or if the system creates unforeseen frictions in a social context.

7.6 Future Work

The theoretical foundation established in this thesis paves the way for several critical avenues of practical implementation and ongoing research. Having identified the epistemic limits of statistical generalization and proposed Social Practice Theory (SPT) as a necessary bridge, future work must focus on the active construction and evaluation of these alternative computational architectures.

Prototyping Hybrid Reasoning Frameworks Future research must prototype the hybrid reasoning frameworks proposed in the discussion. Developers need to construct and test computational models combining machine learning for physical sensor data with symbolic logic derived from SPT. This approach allows researchers to empirically test the efficacy of encoding narrative knowledge as discrete logical units. Specifically, building a functional prototype centered around the Sámi "Sunrise Pasture Assessment" provides a vital case study. This prototype will evaluate how effectively a system can preserve relational coherence alongside quantitative climate forecasts, delivering usable outputs while avoiding the averaging away of localized insights.

Hybrid Interfaces and "Gap Visualization" Future technical work should focus on User Interfaces (UI) that make the "Epistemic Gaps" visible. Algorithmic systems must explicitly flag their epistemic blind spots, such as indicating the presence of snow depth data alongside the absence of ground ice history. This transparency actively supports the herder's epistemic authority. By clearly delineating the boundaries of the machine's computational knowledge, the interface seamlessly hands the final decision-making process over to the human's contextual wisdom.

Participatory Co-Design and Epistemic Sovereignty As this study relied on secondary literature [12, 53], the immediate next step is to ground the "Plan Patterns" in living practice. Addressing the "illusion of inclusion" requires fundamentally shifting the methodology of AI development from extraction to genuine collaboration. Future projects must prioritize participatory action research, establishing Indigenous knowledge holders as primary co-designers. In the context of Sápmi, this involves developing software interfaces and data management protocols directly governed by Sámi herding communities. Subsequent research should evaluate how these collaborative frameworks actively support epistemic sovereignty, ensuring the resulting technology serves the community's self-determined adaptation strategies.

Expanding to Global Indigenous Contexts While this thesis utilizes Sámi reindeer husbandry as a primary proof of concept, the ontological friction between Western computation and Traditional Ecological Knowledge (TEK) applies globally. Future studies should expand this methodology to other Indigenous communities facing rapid environmental transformation. By testing the SPT-based computational framework across diverse ecological and cultural landscapes, researchers can identify universally applicable design principles for decolonial AI. This comparative approach will significantly strengthen the broader applicability of the Sensible AI paradigm.

Policy Integration and Data Governance The transition toward narrative-based, relational AI architectures necessitates updated policy frameworks. Future research should investigate how these specific computational methods align with established Indigenous data sovereignty frameworks, such as the CARE Principles for Indigenous Data Governance. Scholars must explore the legal, regulatory, and institutional mechanisms required to protect formally encoded TEK from the extractive neoliberal practices currently dominating the AI industry. Aligning technical architecture with robust governance will ensure that AI serves as a tool for sustainable self-determination.

7.7 The Paradox of Participation Resolved

A central tension in this thesis was the "Paradox of Participation" [55]: the fear that digitizing Indigenous knowledge might facilitate its extraction or misuse.

This research argues that limitation analysis is a mechanism of protection. By demonstrating the limitations of the formal model, proving that an AI cannot "smell" the snow or "feel" the history of a grazing land or the duty to the reindeer, we establish a hard epistemic boundary. This boundary serves as evidence against "technological solutionism". It asserts that Sámi TEK cannot be fully automated or scraped into a Large Language Model without losing its essential validity. Therefore, the "richness" of TEK revealed by this process is precisely that which remains *sovereign*. This failure analysis confirms that while AI can coordinate the logistics of herding, the wisdom of the practice resides strictly within the human community, resisting all attempts at computational extraction.

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